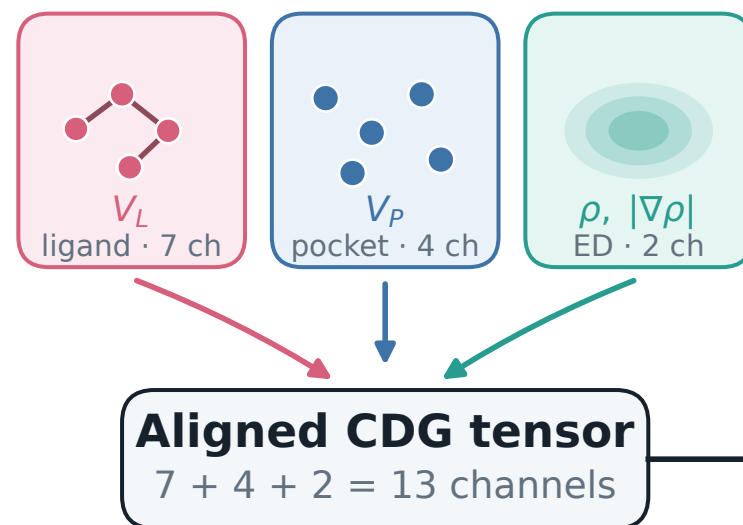
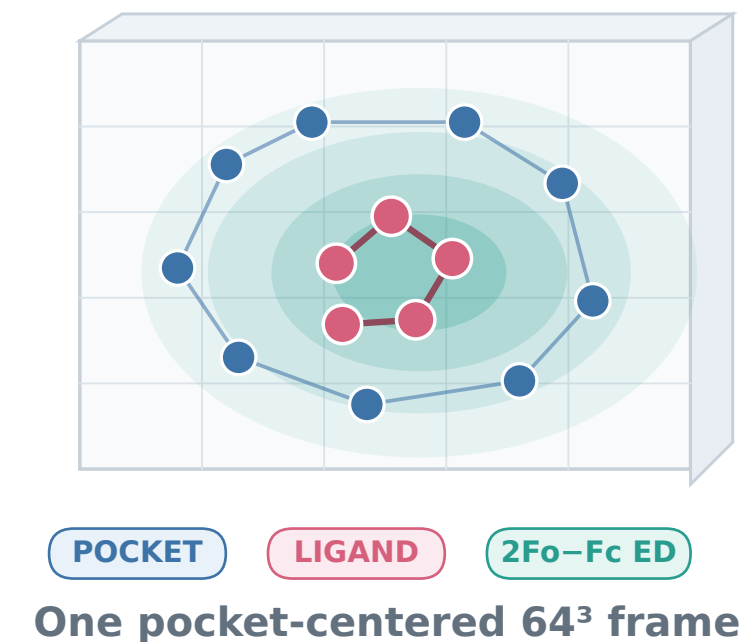


Single-encoder CDG pre-training

Pocket coordinates, ligand coordinates, and experimental electron density are learned jointly in one spatial encoder

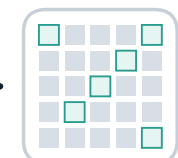
(a) Aligned input data



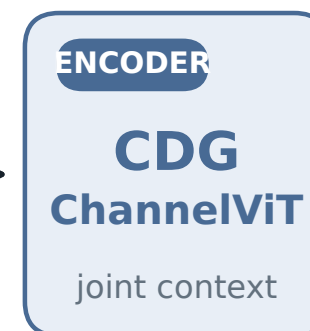
C: coordinates · D: density · G: density gradient

(b) Joint CDG pre-training

Clean CDG x



75% mask



Recon. $\hat{x}$



Masked-only MAE

$$\mathcal{L}(\hat{x}, x)$$

loss on hidden voxels

Reconstruct ligand, pocket, and ED in the same aligned frame
single encoder · single reconstruction target